Electricity Market Clearing Problems: A Reference Formulation

SOC-Based Storage Market Thesis — Reference Note
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Abstract—This note collects, in one consistent notation, the optimization problems that a modern ISO/RTO solves at each layer of a wholesale electricity market: the day-ahead security-constrained unit commitment, the real-time security-constrained economic dispatch, locational marginal pricing, ancillary-service co-optimization, and capacity/resource-adequacy accreditation. It closes by placing the state-of-charge (SOC) based storage participation model of Baldick (2026) within this taxonomy, using the same notation as the accompanying code framework (core/, markets/day_ahead.py).

Index Terms—electricity markets, unit commitment, economic dispatch, locational marginal pricing, ancillary services, resource adequacy, energy storage

I. Two-Settlement Structure

Almost every US and European wholesale market is a two-settlement system. A day-ahead (DA) market clears a financially-binding schedule from forecasts and offers; a real-time (RT) market then physically balances the system every few minutes and settles any deviation from the DA schedule at the real-time price. A capacity / resource-adequacy market sits above both, on a years-ahead horizon – it prices the obligation to be available, but does not dispatch anything itself. Both DA and RT jointly clear energy and ancillary services.

II. Day-Ahead Energy Market

The day-ahead market solves a security-constrained unit commitment (SCUC): which units are on, $u_{i,t} \in \{0,1\}$, and their output $g_{i,t}$, for every interval $t \in \mathcal{T}$.

$$\begin{aligned}\underset{u,g,d}{\text{minimize}} \quad & \sum_t \sum_i \big[c_i(g_{i,t}) + SU_i u_{i,t} + SD_i v_{i,t}\big] \\ & - \sum_t \sum_m u_m(d_{m,t}) \qquad (1)\end{aligned}$$

$$\text{s.t.} \quad \sum_i g_{i,t} + \sum_e (g_{e,t} - p_{e,t}) = \sum_m d_{m,t}, \qquad \forall t \quad \text{(a)}$$

$$u_{i,t} P_i^{\min} \le g_{i,t} \le u_{i,t} P_i^{\max}, \qquad \forall i,t \quad \text{(b)}$$

$$|g_{i,t} - g_{i,t-1}| \le R_i, \qquad \forall i,t \quad \text{(c)}$$

$$\text{min up/down-time logic on } u_{i,t}, \qquad \forall i \quad \text{(d)}$$

$$|f_{\ell,t}| \le \bar{f}_\ell, \quad f_{\ell,t} = \textstyle\sum_n \Phi_{\ell,n} \,\mathrm{inj}_{n,t}, \qquad \forall \ell,t \quad \text{(e)}$$

$$\textstyle\sum_i r_{i,t} \ge R_t^{\text{req}}, \qquad \forall t \quad \text{(f)}$$

$$\text{storage SOC dynamics/limits, if included} \quad \text{(g)}$$

Constraint (a) is the nodal power balance; (b)–(c) gate output by commitment and ramp capability; (d) is the logical minimum-run/down-time restriction; (e) is the DC optimal power flow (DC-OPF) network block; (f)–(g) couple the energy problem to reserves and storage.

III. Real-Time Energy Market

The real-time market re-dispatches continuous output every $\sim$5 minutes against the actual net load via security-constrained economic dispatch (SCED); commitment $u_{i,t}$ is already fixed from the DA solution (or a short-horizon RT commitment for fast-start units).

$$\underset{g}{\text{minimize}} \quad \sum_i c_i(g_{i,t}) \qquad (2)$$

subject to the same balance/capacity/network constraints as (1), evaluated at the real-time net load, with two differences: ramp limits are measured from the last actual dispatch point rather than the DA schedule, and any reserve awarded in the DA market must remain deliverable within its contracted response time.

IV. Locational Marginal Pricing

The locational marginal price (LMP) is the dual of the nodal power-balance constraint (a); it settles both the DA and RT energy markets. Whenever the network is modeled explicitly it decomposes cleanly:

$$\mathrm{LMP}_{n,t} = \lambda_t - \sum_\ell \Phi_{\ell,n}\big(\bar{\mu}_{\ell,t} - \mu_{\ell,t}\big) \qquad (3)$$

where λ_t is the reference-bus energy price (common to every node) and $\bar{\mu}_{\ell,t}, \mu_{\ell,t} \ge 0$ are the congestion duals on line ℓ's flow limit, weighted by the shift factor $\Phi_{\ell,n}$ from ℓ to bus n. A loss component adds a third, often-linearized term when losses are modeled explicitly.

V. Ancillary Services and Reserve Co-Optimization

Ancillary services are co-optimized jointly with energy in both the DA and RT markets: a unit that reserves capacity for a product cannot simultaneously sell that same MW as energy.

$$\underset{g,r}{\text{minimize}} \quad \sum_i c_i(g_{i,t}) + \sum_i \sum_k c_i^k(r_{i,t}^k) \tag{4}$$

with $k \in \{\text{regulation, spinning, non-spinning, ramping}\}$, subject to

$$\begin{aligned}
&\textstyle\sum_i r_{i,t}^k \geq R_t^k, && \forall k,t && \text{(a)}\\
&g_{i,t} + r_{i,t}^{\text{up}} \leq P_i^{\max}, \quad g_{i,t} - r_{i,t}^{\text{down}} \geq P_i^{\min}, && \forall i,t && \text{(b)}\\
&r_{i,t}^k \leq \text{RampRate}_i \times \tau_k, && \forall i,k,t && \text{(c)}\\
&s_{e,t-1} \geq \text{dur}_k\, r_{e,t}^{\text{up}},\\
&\bar{s}_e - s_{e,t-1} \geq \text{dur}_k\, r_{e,t}^{\text{down}}, && \forall e,k,t && \text{(d)}
\end{aligned}$$

Constraint (d) is storage-specific: enough energy (respectively headroom) must already be banked in the SOC to sustain the reserve call for its required duration dur_k. Regulation (seconds–minutes, AGC-driven, near-zero net energy over a cycle) is storage's best operational fit; spinning/non-spinning reserve ($\sim$10-minute response, sustained 30 minutes to a few hours) is comfortable for most storage durations; voltage support needs almost no real energy at all.

VI. Capacity / Resource Adequacy Market

A separate, longer-horizon mechanism: it pays resources to be available during the system's riskiest hours of a future delivery year and does not dispatch anything.

$$\underset{\text{Cap}}{\text{maximize}} \quad \sum_i \Big[\text{VRR}(\text{Cap}_i) - \text{Price}_i \cdot \text{Cap}_i\Big] \tag{5}$$

cleared against an administrative, downward-sloping demand curve (a Variable Resource Requirement, or VRR, curve in PJM-style designs), subject to

$$\textstyle\sum_i \text{ELCC}_i \cdot \text{Cap}_i \geq (1+m) \cdot \text{PeakLoad}, \qquad \text{(a)}$$

where m is the target reserve margin, often set from a Loss-of-Load-Expectation (LOLE) standard of ≤ 0.1 day/year. ELCC_i (Effective Load Carrying Capability) is duration- and penetration-dependent for storage: a 4-hour battery is derated relative to an 8-hour one, and its marginal ELCC falls as more storage of the same duration is added to the system. Crucially, this is not a pass/fail sustained-output test but a probabilistic, Monte-Carlo loss-of-load measure – not a daily discharge requirement.

VII. Storage's SOC-Based Participation Model

The design this framework reproduces (Baldick, 2026) removes the storage owner's in-horizon energy offer entirely: the ISO manages the SOC trajectory directly from the storage's own physical characteristics, priced against a single end-of-horizon bid.

$$\sum_{e,t} \big[\varepsilon_e p_{e,t} + \zeta_e g_{e,t}\big] - \sum_e b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \tag{6}$$

is added to the day-ahead objective (1), subject to

$$\begin{aligned}
&s_{e,t} = \gamma_e s_{e,t-1} + \Delta t\, \alpha_e p_{e,t} - \Delta t\, \beta_e g_{e,t}, && \forall e,t && \text{(a)}\\
&\gamma_e s_{e,t-1} + \Delta t\, \alpha_e p_{e,t} \leq \bar{s}_e, && \forall e,t && \text{(b)}\\
&\gamma_e s_{e,t-1} - \Delta t\, \beta_e g_{e,t} \geq \underline{s}_e, && \forall e,t && \text{(c)}\\
&0 \leq p_{e,t} \leq \bar{p}_e, \quad 0 \leq g_{e,t} \leq \bar{g}_e, && \forall e,t && \text{(d)}
\end{aligned}$$

Constraints (b)–(c) are the tightened SOC bounds: they bind the intra-interval trajectory, not just its endpoints, which is strictly tighter than bounding $s_{e,t}$ alone. The storage owner's opportunity cost is never submitted as a bid – it is recovered by the ISO as the dual of the SOC dynamics (a), $\pi_{e,t}$, propagated backward through the Bellman-like recursion

$$\pi_{e,t} = \gamma_e\, \pi_{e,t+1} - \bar{\nu}_{e,t+1} + \nu_{e,t+1}, \tag{7}$$

terminated by $\pi_{e,T} = w_e(s_{e,T} - \bar{s}_{e,T})$ for a differentiable bid.

VIII. Glossary

Term	Meaning
LMP	Locational marginal price (§IV)
SCUC / SCED	Sec.-constrained unit commitment / economic dispatch
DC-OPF	DC (linearized) optimal power flow
AGC	Automatic Generation Control (regulation signal)
ELCC	Effective Load Carrying Capability (§VI)
VRR	Variable Resource Requirement (sloped capacity curve)
LOLE	Loss-of-Load Expectation (reliability standard)
ISO/RTO	Independent System Operator / Regional Transmission Org.
SOC	State of charge (§VII)
Order 841	FERC's 2018 mandate opening markets to storage

Companion to Code/ROADMAP.md and Literature/KKT_MCP_equilibrium_SOC_storage.pdf – notation matches core/, markets/day_ahead.py.